Case for Case Analysis 1

How Big Data Analytics Improved the Performance of the HAND Company

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Identification of Problems

There are four types of problems that HAND company may have. The problems are operational inefficiencies, data silos and limited insights, client and employee dissatisfaction, and weak cost control and risk management. These are the challenges that can impact the company's operation. Firstly, operational inefficiencies is one of the problems that HAND may have. If the company use the manual system which can be done by human, it can created long wait times and slow process. Like long wait for their approval. If the process can be done by human and if they made a big mistake during the process? It is called human error, and it could delay during the process and could created a long wait time. However, organizations may face some difficulties for track spending and enforce policies which can lead to operational inefficiencies.

Next issues are data silos and limited insights where if the insight is not recovered, it could cover hidden patterns and correlations which can lead to insights get misguided. However, organization face some challenges to keep quality of data and make data driven decisions. (Badman, Kosinski, February 2025) Organizations might have limited data value, data silos can stop organizations from full potential of their data assets. Organizations face some difficulties to build comprehensive datasets, so it might have to get it done by big data analytics and machine learning activities. (Badman, Kosinski, February 2025)

What happen if the client or employee wait for several days for their pending approval and keep waiting. If they had a long wait and getting not patient, it can lead to dissatisfaction. Also, if there are errors that happening during the process, it could upset the client and employee because if errors keep happening, it could keep delay as sometimes they might need to redo the process or had to fix the process to prevent from potential delays. This could affect the company operation and satisfaction for client and employee.

If the accountants are working without using accounting information system, HAND may get some difficulties to analyze numerous financial and non-financial data. If the risk management is limited, could have high cost and budget inefficiencies. (Alexandra, June 2025) Organizations could get misaligned priorities and suboptimal outcomes if the cost is higher and not efficiencies for budget. (Alexandra, June 2025)

Decision Criteria

Data Accuracy	• Correct inventory tracking, reporting, and client information • Not making mistakes on order, billing, or client's interactions • Helping operational to be efficiency, could have significant analytics and insights
Real-time Accessibility	• Up to date for number of stocks, orders, or service request • Able to response promptly and make workflows efficiently • Supports proactive problem-solving
Client Satisfaction	• Received the correct products or services on time • Receiving no damaged products • Less errors in orders or informed for keeping update if anything happens • Personalized service helps to accurate client data • Higher trust and loyalty

Global Integration	• Consistently in data across international operations • Supports financial reporting and strategic planning • Compliance with global standards and local regulations
Cost-Effectiveness	• Minimizes waste due to order mistakes or inventory error • Reduces labor and administrative cost by automating reporting and analytics • Support better resource allocation

Analysis

Based on the identification of problems based on operational inefficiencies, data silos and limited insights, client and employee dissatisfaction and weak cost control and risk management. It can affect the company's reputation if there are any if identification of problems exists. The first problems that HAND's company could face many difficulties prior to ERP integration and adopting big data analytics exists. However, if organizations use manual reporting like paper-based expense could lead to operational inefficiencies. If anything, that can lead to not efficiency for operation, it can lower trust and misguided strategic decisions. (Achievelt) If more errors, increasing operational issues, or keep delaying can lead to confusing and complexity. (Achievelt) It relates to HAND's issues when the organization making more confusion, it can slow down decision-making. It is complicated. Organization manager could lose their confidence. Another example like further increase waits times for their approval and reimbursement. It can be dissatisfied with employee and lower productivity.

Another problem are data silos and limited insights. Data can implode when organization focus more on financial data. Ignoring non-financial insights like performance, travel, and expenses. If the organization tries to overfit the density model, this can lead to infer incorrectly. (Zhao, Papernot, Anderson, Gal, July 2024) If organization made a mistake that can infer incorrectly, it could make bad business decisions. It would be the biggest mistake for the business. Like if HANDS tries to overfits the data model, predictions result may not shown accurate. It can be lack of management's ability to detect risks or finding opportunities.

Let say if client or employee waiting for long time, like weeks, it could decrease in overall productivity and not efficiency. This could lead to experience burnout, decreased job satisfaction and higher turnover rate. (Mazzucco, March 2025) More client and employee dissatisfaction could disrupt the continuity of service, further increase waits times. (Mazzucco, March 2025) HANDS employee face errors, delays, and unreliable data can lead to unsupported and overwhelmed. As mentioned by Mazzucco, if HANDS employee gets unsupported and overwhelmed, it can have burnout because of difficult to catch it up. High turnover could increase the training costs and lower the company's ability for their operations. If the company gets operational disruption like getting not accurate or overfitted insights, can lead to delay service and slow response time. It can upset the clients, affect the satisfaction, and brand reputation.

Lacking integrated system, manager had lack of visibility into real-time spending can cause week cost control and risk management. Previous identified risk can evolve, and new risk can appear. (Tuvsud, October 2023) If HANDS experience with data inaccuracy or operational delays, these are current risk. It can appear as company expands globally. If the new risk keeps

adding, it may change in regulatory, supply chain disruption, or problems with technology may arise. As the result. HANDS must monitor the risk and acting fast by using adaptive strategies to keep it resilient.

Alternatives

Alternatives	How does it work?	Benefits	Limitations
Manual Reporting with Spreadsheet Systems	Can be done by employee who use either spreadsheet or logs for tracking expenses and project data manually.	▪ Low cost ▪ Customizable formatting ▪ Quick to implement	▪ Human Error Prone ▪ Slow Process ▪ Data Security Concerns ▪ Visibility issues
Traditional On-Premises ERP Systems	Deploying ERP software allows for the greatest level of control, offering a high degree of control and customization.	▪ High customization ▪ High Integration ▪ Mature system functionality	▪ High Cost ▪ Consume more time ▪ Difficult jobs for any organization
Outsourcing Expense Management	Hire third party services to help handle expense reporting and compliance	▪ Reduce internal workload ▪ Speed up the processes. ▪ Access to specialized knowledge and tools	▪ Less Control ▪ Data Security Risk
Cloud-Based ERP Solutions	Solutions to all the difficulties encountered by conventional ERP systems.	▪ Flexibility ▪ Improves overall efficiency and cost-effectiveness	▪ Data Privacy Concerns ▪ Big Issues with storing sensitive information off-site

Decisions / Recommendations

Based on the recommendations, it would be better for the HANDS company to have ERP systems like Helios, ERP system, and big data analytics. The more they use, it helps to have potential future growth, less cost, more efficiency, provide services on time and reduce risk. ERP systems like Helios is recommended because it able to identify the issues with the organizations due to manual inefficiencies, data loss, and weak cost visibility. Helios helps to reduce cost and increase compliance. It provides comprehensive, scalable, and secure approach which gains more efficiency on both short-term and long-term strategic capability. As an example, it can calculate and estimate the mileage for their business with pressing the button only one time. It definitely reduces the hassle of manual tabluation. Helios system is like SAP and Oracle, it making sure to stay contunuity without disruption for workflow. By staying on the track and not fallen behind

would helps if they use Helios system because Helios provides real-time data analysis. It can reduce the hassle of manual tabluation.

Using big data analysis is a great idea to stay on track because it helps to gains the business value and less risk for organizations enterprise in borth short and long term. Big data analysis definitely helps with effective cost containment and enterprise control. It provides quantitative analyses which would be helpful for enterprise manager because of getting a reference control model for data model. It creates competitive advantage. It is good for the HANDS company for their future growth in data driven environment. Big data analytics is part of strategic advisors which helps the accountants enables their transitions from manual system. It allows to stay on track by conduct project costing, workforce analysis, and predictive modeling.

Implentation

1. **ERP Integration:**
    - Using the updated application technology and integration with back-end ERP like Helios.
    - Like SAP and Oracle system, Helios can integrate with all legacy system.
    - Using Helios, latest application technology and inegration with back-end ERP. allows seamless integration process with ERP system so it will be smooth process.
2. **Centralized Data Security:**
    - To keep the data information safe, it will have centralized and encrypted database in the same secured server.
    - All user information are being secured and save automaticlly with an additianal layer of security.
3. **Quality Control and Compliance:**
    - Using Helios would increase in compliance and direct expence saving. And able to reduce in transation and approval cost.
    - Would increase in overall productivity and efficiency.
    - To enforce company and expecting policies, company is better to use built-in compliance
    - Schedule audits and inspection to check to see if any issues, errors, or fraud. This will prevents from getting worse. (Technia)
4. **Training:**
    - Provide mandory training for employee to familiarize with harnessing the power of data analytics to make data driven insights.
    - To maintain competitive advantage over other organizations, organizations need to know about project costing and workforce analysis well.
    - Need to know how to capture effectively, manage, and share the information across the entire corporation to ensure the right decisions in the right time.
5. **Data Quality and Tools:**
    - Quality of data gathering is important for consideration based on two other factors
    - Using relevant data analytics tools and techniques:
        - Understand, drae insights, and storytelling from the data
        - Can incorporate non-financial data

    - Perform advanced analytics processes
- Pinpoint opportunities and potential risks for business improvement and future expansion considerations which can be done by big data analytics.

Question and Answer

1. The competitive advantages which big data analytics can create for an organization could be based on data quality, trends, patterns, and correlations, risk management, real-time insights, enterprise control and cost containment, and sustainablea competitive advantage. Firstly, the data quality, in order to collect and process information from multiple sources without disruption is to improve data quality. Secondly, to help to have better insight and business decisions is to reveal trends, patterns and correlations. Thirdly, if the organizations want to make it faster, more accurate decision-making and going smooth is to authorize real-time insights. Then, in order to make profitability better, it can strenghtens enterprise control and cost containment. Finally, providing sustainable competive advance through data-driven strategies.

2. Accountants can leveraging big data analytics to transition from traditional financial reporting to strategic decision-making roles within organizations by generate data-driven insights, manage project costing and workforce analyses to improve resource allocation. Using non-financial data can be performed advanced analytics proesses. Making the right decision and using long-term planning that can be done for support management. Maintaining a competitive edge over other organizations to identify the first sign or early sign of risk and opportunities early.

3. The adventages for organizations face in ensuring data quality and accessibility across departments are using high quality data make get accurate results for insights and decisions, sharing data accress by supporting cross-department collaboration, imrproving compliance and transparency and have seamless integration like SAP and Oracle which across the legacy systems. However, the disadvantages are more risky for data security and privacy from centralized databases, using vendor depency can expose to organizations lead to breaches or system failures, can have incosistent or inaccurate data which can slow processes due to poor governance, and have operational and reputaitonal risk if vulnerabilities in integration. The challenges can impact the success of digital transformation initiatives which drives efficiency, compliance, and better decision-making which is the positive sides. However, the cons which is the negative side could be lack of security can lose their trust, stall adoption, and have financial or reputational damage.

4. HAND enterprise solutions used big data analytics and ERP integration to improve project performance and client satisfaction across its global operators are ERP integration, centralized data, efficiency gains, employee and client satisfaction, project performance, and strategic insight. Using ERP integration allows to reduce silos by having seamless integration of Helios. To protect data by encrypted database which improves security, compliance, and real-time reporting which is centralized data. Showing efficiency gains which are 40% decrease in approval cycles, 60% increase in compliance, 55% reduction in transaction and approval cost. Employee and client satisfaction shows faster process in expense reimbursements, easier claim submissions which improve morale, accurate reporting, more trust, and better service quality. To have better cost control, using analytics of labor, travel, and accomodation as a projecy

performance. To use better tools for cost control and performance evaluation which use for strategic insights which it is provided by managers for better data model.

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